PROPOSAL

Develop an Intelligent Chatbot

Department of Computer Science

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1. Introduction

The increasing complexity of the lives of students in higher education institutions has raised many issues concerning mental health and well-being, career planning, and practical problems such as student housing, especially in international students finding their way in a new environment. Although there are resources established within the educational institution – the University of East Anglia (UEA) in the present case – lack of access and awareness regarding these resources is the major problem for quite a few students. In order to overcome this problem, the current paper suggests creating an AI chatbot which will provide assistance on the following issues: personal well-being, career planning, student accommodation issues of international students, and living in Norwich in general. The rest of the current paper will focus on reviewing relevant literature, defining the goals and objectives of this research, explaining methodology, as well as discussing ethics.

2. Background and Project Description

2.1 Background

Chatbots are machine agents that engage in communication with humans using natural language. Alan Turing in 1950 proposed the Turing Test ("Can machines think?"), and it was at that time that the idea of a chatbot was popularized. The first known chatbot was Eliza, developed in 1966, whose purpose was to act as a psychotherapist returning the user utterances in a question form. Over time the main focus in chatbot design moved from the perfect imitation of humans to the development of helpful tools able to support the users in their tasks with a more natural interaction achieved through the use of natural language and using machine learning methods to train the chatbots to understand user inputs.

The idea for the proposed AI chatbot came about because students at the University of East Anglia (UEA) have trouble getting reliable and up-to-date information. There are platforms like SIZ and Blackboard, but they often have to be navigated through many links, which takes a lot of time. One big worry is that students have a hard time finding safe places to live because many online sources aren't reliable. Additionally, students who live far from campus often need quick info about transportation. International students may also have trouble finding grocery stores that fit their cultural needs. Students are more and more likely to want quick, direct answers in one interaction instead of having to search through complicated systems.

Existing support services are limited by time. Students often need help at any hour, especially late at night, when anxiety or loneliness can increase. Many students feel uneasy discussing sensitive topics like well-being or disabilities, making a private chatbot a better option. The chatbot aims to offer a reliable, accessible solution that provides instant, personalised, and confidential support to improve the overall student experience.

2.2 Problem Statement

Although the technology of chatbot application is rapidly becoming mainstream, it remains generalised, and fails to cater for the particular needs of university students as a population group. Although some universities have implemented the idea of chatbot application to offer student services, such an approach remains limited since it addresses the administrative needs only, but fails to provide any meaningful assistance regarding issues related to mental health, career development, and housing, among others. Moreover, there are significant obstacles for many students – especially international ones – to use student services offered by universities due to lack of awareness, social stigma and the traditional structure of these services, which are office based. Currently available chatbots do not allow users to personalise their experience and do not contain elements of emotional intelligence.

2.3 Project Description

The proposed system is a web-based AI powered chatbot that will operate as a digital assistant for the students of the University of East Anglia, using NLP technology to interact with users in a meaningful and context-sensitive manner through the web interface of the application. The chatbot will assist students in four main spheres – personal welfare, career planning, accommodation issues, and general student life tips in Norwich, giving the right information and connecting users with the relevant services provided by UEA where needed. The scope of this project will remain confined to developing and testing a prototype for the use of the University of East Anglia.

3. Aim and Objectives

3.1 Aim

The main aim of the project is to build a chatbot for UEA students that will provide assistance in various spheres of interest including health & well-being, careers advice, accommodation advice, and orientation to the city of Norwich where the University of East Anglia is located.

3.2 Research Objectives

To fulfil this aim, the following specific objectives have been formulated:

- To review the current literature on intelligent chatbot systems and conversational AI.
- To analyse the requirements (functional and non-functional) of the proposed chatbot system.
- To develop a complete architecture of an intelligent chatbot using NLP and dialogue management.
- To design the chatbot system using appropriate tools, frameworks and programming languages.
- To test and evaluate the performance of the developed chatbot with standard metrics.

- To review the development process and to suggest directions for future improvements.

4. Critical Review of Related Literature

4.1 Chatbots for Mental Health and Student Well Being

Chatbots developed using artificial intelligence (AI) are emerging in the field of psychology. Currently, there are two chatbots that have addressed anxiety and depressive symptoms, Woebot and Tess. Woebot is a chatbot based on the cognitive behavioral approach with evidence for the reduction of anxiety and depressive symptoms in students during a follow-up after 2 weeks. Fulmer reported a reduction in depressive and anxiety symptoms in college students using Tess, a chatbot that provides support and psychoeducation through an integrative approach. Although the research completed by Fulmer and Fitzpatrick reported decreased depressive and anxiety symptoms in college students, these studies were performed in the United States. To the best of our knowledge, there are no studies on chatbots used for addressing mental health disorders.

4.2 Chatbots in Higher Education

Another research of chatbot, Deakin University's virtual assistant, "Genie", represents an example of the application of Chatbots in education. The platform, presented through an iOS and Android mobile application, utilises artificial intelligence, voice recognition and predictive analytics. It is integrated with existing systems, including the University's learning management platform, digital library and IBM's Watson question-answering system (Scheepers, Lacity, & Willcocks, 2018). However, there are certain drawbacks of "Genie" are contextual limits , like AI assistants still struggling with complex query and also its difficult to maintain such an advanced personalized system. It also has a gap of monitoring student data, including geolocation for finding classrooms, and mood tracking, which raises privacy and ethical considerations.

4.3 Ethical Challenges in Educational Chatbots

Ethical challenges associated with the use of chatbots in education is the potential for the technology to replace human interaction and expertise. This is particularly concerning in fields such as counseling and mental health, where students may seek emotional support from chatbots instead of trained professionals. Another challenge is the potential for bias in chatbots. AI systems are only as unbiased as the data they are trained on. If the data used to train chatbots are biased, then the chatbot's responses may also be biased. This could result in unfair assessment outcomes and could potentially perpetuate discrimination and inequality in the education field.

Chatbots have emerged as a promising educational tool, with the potential to enhance the learning experience by providing personalized and immediate feedback to students. However, the use of chatbots in the educational field also raises ethical challenges that need to be addressed.

4.4 Identified Research Gaps

Lack of Personalisation : Present-day chatbot applications have several limitations in addressing complex scenarios of student advising by providing standardized rather than personalized.

Limited Scope : A significant number of chatbots developed for universities are designed for answering certain administrative or academic questions only; thus, there are no holistic solutions aimed at students' needs beyond learning activities.

Problems with Datasets and Usability : Some problems associated with chatbots in the field include limited datasets and issues related to usability. Additionally, there are not enough real-world cases for validation of proposed applications.

Barriers to Mental Health Support : Barriers for students to receive needed mental health services include stigma, unaffordable treatment, lack of providers, and waiting times, which can be addressed using AI-based chatbots.

5. Methodology and Methods

From the literature review of chatbots, examples such as Woebot, Tess and Genie have the capabilities to use the chatbot in educational and well-being, but it has some limitations, such as restriction of context, privacy issues, bias when responding and the inability to respond to complex questions

5.1 System Architecture Methodology

The User Interface will be created by HTML/CSS and JavaScript, and its framework. The backend of the chatbot is implemented in Python, with the FASTAPI framework handling all API calls between the chatbot interface and the AI model. The model will use the Large Language Model(LLM) to generate natural-sounding human responses to queries regarding the student's well-being, house searching, career planning, questions related to campus and any other questions they may have about the city. Additionally, the chatbot response will be generated using a retrieval-augmented generation(RAG) process, allowing the chatbot to retrieve up-to-date information from a trusted knowledge base before generating a response..

5.2 Data Collection and Preparation

The critical part of the chatbot is to gather the relevant data and accurate data to train and feed into the RAG knowledge base. The following methods will be used to collect the data.

- UEA official website - Publicly available information from the UEA website will be scraped and collected, including details about student services, well-being support, accommodation, and campus facilities.
- SIZ and Blackboard Platforms - Relevant student support information available on these platforms will be gathered and structured into the knowledge base

- Surveys and Questionnaires - An online survey will be distributed to UEA students to understand their most frequently asked questions, needs, and expectations from the chatbot system.
- Public Datasets - Publicly available datasets related to student well-being, housing prices in Norwich, and transportation information will be collected and used to enrich the chatbot's knowledge base.
- Web Scraping - Python-based web scraping tools such as BeautifulSoup and Scrapy will be used to extract up-to-date information from trusted online sources such as housing websites, transport websites, and local grocery store listings relevant to international students.

5.3 Requirements Analysis: MoSCoW Method

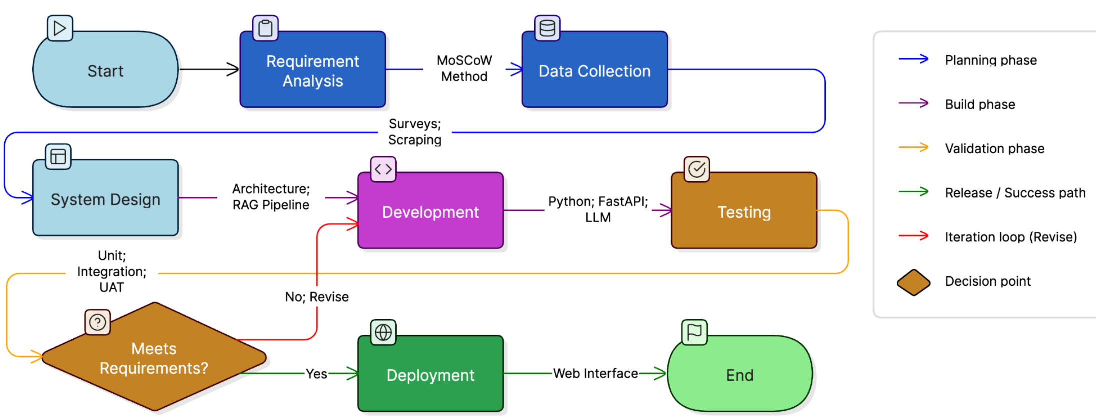
MUST HAVE	
1.	The chatbot needs to provide correct and suitable replies to student wellbeing questions.
2.	The chatbot needs to provide students with accurate information about housing and accommodation.
3.	The chatbot has to be available 24/7 to give instant support to UEA students.
4.	It should be able to answer career related questions and provide suitable guidance to students.
5.	The chatbot needs to take natural language input and process it with NLP and LLM to understand it.
6.	The chatbot should be available through a web interface developed with HTML, CSS and JavaScript.
7.	The chatbot should be built using RAG approach so that it can fetch accurate and latest information.
8.	The system should be stored and manage data securely using an appropriate database.

SHOULD HAVE	
1.	Provide information about the city of Norwich including transport and local services.
2.	Provide information about UEA platforms such as Blackboard and SIZ.
3.	Take care of follow-up questions in the conversation.
4.	The chatbot should be able to respond to the queries related to career and give apt guidance to the students.
5.	Provide grocery store information for students.
6.	Log the conversation history for future improvement.
COULD HAVE	
1.	Support multiple languages to better serve international students.
2.	Send notifications for assignment due dates and important events.
3.	Add a feedback option at the end of every conversation.
4.	Offer mental health resources and direct students to professional support services .
5.	Add an admin dashboard for system administration.
WOULD NOT HAVE	
1.	Replace professional well-being counselors or provide medical diagnoses.

2.	Execute any financial transactions/payments.
3.	Keep sensitive personal data longer than is strictly necessary for the session.
4.	Be available as a mobile app in this version.
5.	Live chat with university staff in real time.

5.4 Methodology Design

The proposed intelligent chatbot system is developed based on an Agile development methodology which allows the system to be iteratively developed, tested and improved with continuous feedback. The development process consists of seven major phases as shown in the flowchart below starting from requirement analysis and data collection, system design, development and testing. Following the testing phase, a decision point is reached to determine if the targeted performance standards have been achieved. If not, it is revised and re-tested. The final stage is deployment, when the chatbot becomes available to UEA students through a web-based interface.



6. Analysis of Risks

Technical Risk	Mitigation Strategy

NLTK and spaCy can potentially misunderstand informal expressions, slang, or sophisticated questions from students.	It should be trained on a variety of representative student questions and tested iteratively during the development phase.
A lack of domain-specific data can make the system fail at providing appropriate responses to students' questions about UEA.	One should supplement existing data with hand-curated datasets and keep improving the model based on user feedback.
The system will take long to respond or crash under many concurrent requests from users.	The codebase should be optimized in terms of performance, load testing should be conducted before deployment, and the web infrastructure should be scalable.
Sensible processing and storage of personal data from students can cause security issues.	The system should be developed in line with GDPR requirements, no personal data should be stored without consent, and all transmissions should be encrypted.
In case the training data is not diverse or representative enough, the chatbot might respond inappropriately or generate biased messages.	Such outcomes should be prevented by conducting regular audits, collecting diverse training data, and implementing content filters.

7. Project Work Plan and Gantt Chart

7.1 Project Phases

The project will be executed across four principal phases, spanning sixteen weeks from project commencement to dissertation submission:

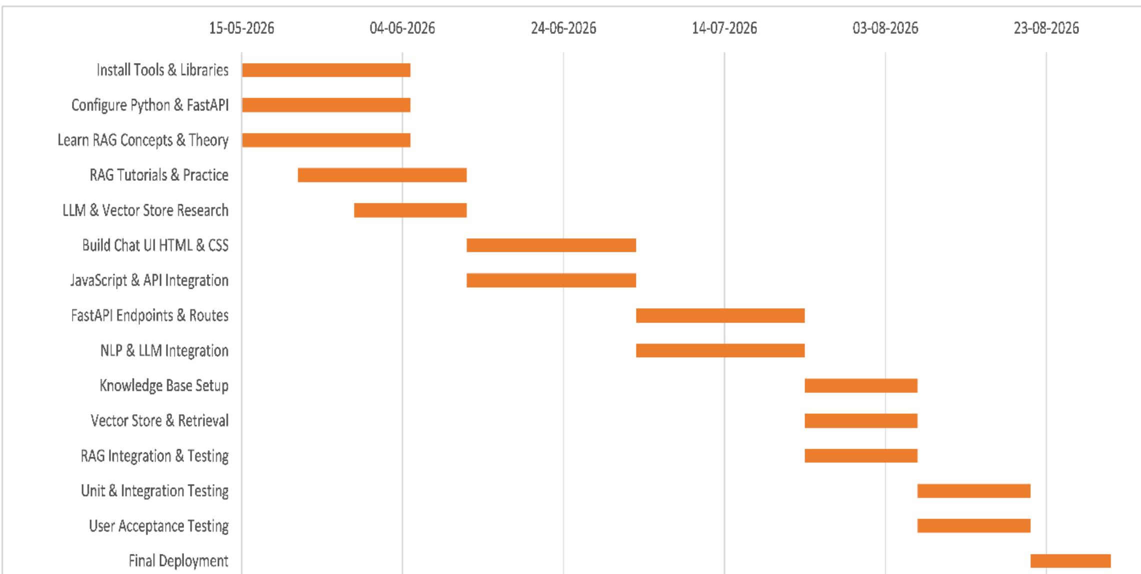
Phase 1 - Environment Setup & Research : In this phase, we concentrate on creating the technical base for our project by installing necessary tools, configuring Python and FastAPI, studying RAG theories and concepts, working on RAG tutorials and practices, and researching LLM and vector stores technologies.

Phase 2 - Frontend & Backend Development : The key components of this stage include the construction of the chatbot interface using HTML and CSS coding, programming with JavaScript and API, designing the FastAPI endpoints and routes, as well as incorporating NLP and LLM into the process.

Phase 3 - RAG Implementation & Knowledge Base: This step concentrates on the capabilities of intelligent retrieval of the chatbot and includes knowledge base configuration, setting up vector stores, and setting up the retrieval process. This will also include the implementation and testing of the RAG pipeline for accurate responses.

Phase 4 - Testing & Development : Phase four involves everything related to testing and deployment, such as unit and integration testing, User Acceptance Testing using students from UEA, and deployment of the chatbot.

7.2 Gantt Chart



8. Conclusion

In summary, this proposal outlines the implementation plan of an AI-enabled chatbot to help address the increasing demand for easy access to and personalised assistance from University of East Anglia, across four main topics: personal well-being, career guidance, housing advice, and general information about living in Norwich. By using large language models, retrieval-augmented generation techniques, and NLP in Python, alongside FastAPI for its back-end infrastructure, this project aims to fill the current gap between university students and their options by providing round-the-clock support, an impartial attitude, and context-sensitive advice in the form of a bespoke AI assistant to help them navigate the complexities of student life. Although the scope of the project does not include providing replacement mental health and career counselling services, this effort may contribute to enriching the student experience of UEA students via the application of conversational AI technologies. Ultimately, the successful completion and assessment of this project can lead to positive outcomes both for the target population and for the discussion of AI in higher education settings in general.

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